

Brivalemer Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-450/C

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The control strategy links critical quality attributes to critical process parameters identified during development.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Statistical evaluation of batch release data demonstrates process capability well within specification limits. No changes to the approved analytical procedures are proposed in this submission.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend

uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Cleaning verification swab results were below the health-based exposure limits derived for the product.

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Control of Excipients (3.2.P.4)

For the commercial formulation F-450/C, Tablettose 80 (Lactose monohydrate, diluent) is used.

Grade selection and control follow the principles of ICH Q6A, ICH Q8(R2) and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Selection Justification

The selection of Tablettose 80 is justified by reduced lot-to-lot variability reported by the qualified supplier observed during development studies.

Approved Drug Product Specification

The current approved commercial specification (SPEC-EX-4609 Rev 01) lists Lactose monohydrate as Pharmatose 200M.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The control strategy links critical quality attributes to critical process parameters identified during development.

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Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

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